

Sebastian Numpaque

Astronomer

[Mail](#) | [GitHub](#) | [ORCID](#) | [LinkedIn](#)

About Me

Undergraduate student in Astronomy at the University of Antioquia. Young Researcher Intern in the FAcOm and SEAP research groups, focusing on exoplanetary science. Coordinator and Co-founder of the Astronomy Research Seedbed for students. Organizational Manager at Astrodidaxi Foundation for Astronomy Education. I have made contributions in the design and development of scientific and computational tools, as well as in the publication of scientific papers and organization of science outreach activities.

Professional Experience

Young Researcher Intern - FAcOm & SEAP Groups, University of Antioquia

Medellín, Colombia | 2024 – Present

- Developed scientific Python packages for exoplanetary research.
- Managed open-source projects and collaborative development on GitHub.
- Implemented software testing and documentation workflows.
- Co-developer of the package *Pryngles* - <https://github.com/seap-udea/pryngles>

Organizational Manager - Astrodidaxi Foundation for Astronomy Education

Boyacá, Colombia | 2023 – Present

- Research project management
- Impact of local communities in Boyacá
- Planning and execution of academic and outreach events
- Coordination of initiatives to strengthen the foundation's visibility and impact.

RECA Mentorship Program, Colombian Astronomy Students Network (RECA) | 2025

- Management of the Call for Applications

- Planning of Panels and Activities
- Communication and Promotion
- Organizing Committee Member, RECA Mentorship Node

Featured Projects

Pryngles

Co-developed a Python package for simulating exoplanet transit light curves, supporting research in exoplanetary science.

[Project Link](#)

Astrophysics Research Training Program

Co-founded and coordinates a training program in astrophysics research for undergraduate students, focusing on computational, communication, and scientific skills.

Education

Astronomy Undergraduate Programme - University of Antioquia

Medellín, Colombia | 2021 - Present

- Recognition Award for Best Advanced Student in Astronomy, 2025

Complementary Courses

- Exoplanets | University of Geneva - Coursera, 2024
- Data Mining | Pontificia Universidad Católica de Chile - Coursera, 2021
- Introduction to AI | DeepLearning.AI - Coursera, 2021
- Formulación de Proyectos de Investigación | Vicerrectoría de Investigación, Universidad de Antioquia, 2025

Skills

- Software Development (Python)
- Object-Oriented Programming
- Collaborative Development (Git, GitHub)
- API Documentation (ReadTheDocs, Sphinx)
- Linux System Management
- Statistics & Data Science

- Physical-Mathematical Modelling & Simulation
- Scientific Research & Outreach
- Project Design & Management

Offered Workshops and Courses

- Explorando Mundos Distantes: Detección de Exoplanetas y la Zona Habitable
 - AstrodidaXis Network Foundation | 2024
 - 27° Astronomy Fest, Villa de Leyva, Colombia | 2024
 - Luis Carlos Galán Sarmiento School, Villavicencio, Colombia | 2024
- AstrodidaXis Young Course on Astrophysics | 2025

Speaker

- **VIII Congreso Colombiano de Astronomía y Astrofísica (CoCoA) 2024**
A Python Package for Modelling the Photometric Signatures of Exoplanetary Rings/Satellites
- **III Congreso Internacional de Educación Ambiental Comunitaria 2024**
Explorando la Contribución de la Astronomía a los Objetivos de Desarrollo Sostenible y el Empoderamiento Comunitario en Boyacá

Assistant

- VII Congreso Colombiano de Astronomía y Astrofísica (CoCoA) | 2022
- II Congreso Iberoamericano de Estudiantes de Física (CIEF) | 2023

Organizer Committee Member

- II & III Workshop on Astronomy Education. OAE - Colombia, AstrodidaXis, UPTC | 2023, 2025

Additional Publications

- Artificial Intelligence in the Teaching of Natural Sciences on the Threshold of the Fifth Industrial Revolution. Explainable AI for Education: Recent Trends and Challenges. 2024 | Book chapter. DOI: [10.1007/978-3-031-72410-7_9](https://doi.org/10.1007/978-3-031-72410-7_9)
- Aportes de la Astronomía y su enseñanza a los Objetivos de Desarrollo Sostenible, caso AstrodidaXis (Paper Submitted). Góndola, Enseñanza y Aprendizaje de las Ciencias ISSN: 2665-3303 | 2025

- Explorando Mundos Distantes: Taller en Detección de Exoplanetas. Circular Astronómica No. 999, Red de Astronomía de Colombia (RAC) | 2024. [PDF](#)

Languages

Language	Level
Spanish	Native
English	B2

Certifications

To see all the current certifications, please see [Certifications](#)

